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ABSTRACT

In order to establish a long-term relationship between a robot and a child, robots need to learn from the environment, adapt to specific user needs and display behaviours and roles accordingly. Literature shows that certain robot behaviours could negatively impact child's learning and performance [17]. Therefore, the purpose of the present study is to not only understand teacher's opinion on the existing effective social behaviours and roles but also to understand novel behaviours that can positively influence children performance in a language learning setting. In this paper, we present our results based on interviews conducted with 8 language teachers to get their opinion on how a robot can efficiently perform behaviour adaptation to influence learning and achieve long-term engagement. We also present results on future directions extracted from the interviews with teachers.

Author Keywords

Social and Adaptive Robots; Language learning; Children Robot Interaction; Long-term Engagement.

ACM Classification Keywords

H.5.m. Information Interfaces and Presentation (e.g. HCI): Miscellaneous

INTRODUCTION

In recent past years, research in Human-Robot Interaction (HRI) and Social Robotics (SR) has emphasised immensely on the need and benefits of social robots that can adapt to specific user needs. Researchers have conjectured these benefits through conducting a series of studies with non-adaptive robots. They have showed that due to the lack of flexibility of a non-adaptive robot in its response, specifically to the behaviour of humans. People tend to loose interest and the impact of such robots decreases with time [16]. Researchers have also pointed out benefits through both qualitative [10] and quantitative [29] evaluations.

Social Adaptive Robots (SARs) have been defined as autonomous robots/agents that can learn from a social environment and adapt to user behaviour accordingly. Therefore, in order to implement these SARs, it is important to understand what adaptable social behaviours and roles displayed by the robot are acceptable to humans in different domains and environments. Researchers have inferred that there is an uncertainty on what is meant by adaptive agent behaviour in different domains. They are trying to understand various appropriate adaptive behaviours that can lead towards efficient learning. Dautenhahn et. al., [5] and Mahani et. al., [22] have conducted interviews with potential end users to derive social roles. However, we argue that interview answers were based on the hypothetical knowledge on robots. In literature, we also find general behaviours such as understanding and reacting to emotions, playing different social roles [9]. Researchers previously have tested these behaviours both quantitatively and qualitatively, however, recently, Huber and colleagues [14] have critiqued that social robots need to extend on the stereotypical role behaviours in order to achieve long-term acceptance. They followed different steps (application scenarios identification through Interviews, listing different adaptive roles through qualitative analysis, frequency of adaptive roles mentioned differently by elderly adults in home and care centres, classification of adaptive roles according to age and demography) to identify various adaptive behavioural roles for elder adults living at home or in care facilities.

We understand that the view of students is important in the pedagogical relationship between an embodied tutor and students. Obaid et. al., [25] conducted an exploratory study on children's contributions to the design of a robotic teaching assistant. He compared children and interaction design students design. He showed that robot designs are influenced by individuals knowledge of robots. Serholt et al., 2014 [30] has also interviewed students to get information on their attitudes towards the possible future of social robots in education. However, the role of teachers must not be underestimated. According to [2], it is described that the teachers play an integral role in how children learn using aids, devices in class and perceive information. Even though devices can function independently, the teacher must direct the flow of the session/curriculum. Therefore, HRI researchers have given importance to understanding the view of teachers on understanding the possible interactions for robots.

Researchers have also conducted studies through interviewing teachers at the schools in order to understand the accep-

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tance and possible experimental methods for children robot interaction at schools. Serholt and colleagues [31] have conducted interviews with eight teachers from four different European countries on their views about the use of emphatic robotic tutors in the classrooms. They questioned teachers about the plausibility of using robots, what roles should a robot play, and what are the overall positives and negatives of having robots in education. They have found that robots can be a disruptive technology as it can be an extra administrative task to allocate personalised use of robots fairly among children. Teachers also pointed that in case a robot is programmed autonomously, it can be helpful in managing group work and can also promote independent learning. In addition, robots can also guide and motivate students through several learning tasks. As [25] showed previous knowledge had an effect on robotic designs, it can be argued that teachers responded to the questions based on the prior hypothetical knowledge of robots or on knowledge collected through science fiction movies. Therefore, some responses can be categorised as biased.

Most recently, Kory and colleagues [19] have shown results contrary to findings discussed by [31]. They have also argued that due to the hypothetical knowledge of teachers on social robots, their observations and opinions in some case were not supported. [19] deployed an autonomous Tega robot in the classroom for 2 months and then asked view of teachers. The Tega robot was not only capable of playing with a child but also administer curious children waiting for their turns. Results showed that teachers were pleasantly surprised with how non-disruptive the whole experience was and how much children were positively engaged with the robot. Keeping [19] study in view, we believe the next steps are to inquire teachers about behaviour and roles for adaptive robots through first enabling them to interact with the robot and understand its capabilities. In addition, we need to find appropriate behaviours for robots to help with language learning. In this paper, we present a study conducted with teachers from different schools in a urban city. The focus of our study is to understand about the most appropriate and effective children robot interactions in language learning education. In order to avoid the influence of prior knowledge of teachers on results, the teachers interacted with a humanoid robot and they were informed about all of its possible and probable capabilities. As discussed by [14], we also believe in order to implement adaptive robots to facilitate long-term engagement, it is very important to involve teachers to guide and help us design appropriate and effective behaviours for robots. In addition, recently, Kennedy and colleagues [17] showed that the robot who over tried negatively influenced child's learning. Therefore, we need to careful and should make informed decisions before implementing behaviour adaption strategies and features for social robots who are going to interact with children.

EDUCATIONAL ROBOTS FOR LANGUAGE LEARNING

The use of robots to help children in language education has attracted an enormous amount of attention globally. One of the reasons is that robots possessing an embodiment offer to play a variety of social roles that can help with learning. Researchers have recently used these agents to teach children

about science, mathematics, languages and robotics. [16] conducted a field trial with a humanoid robot, Robovie, as a partner to help children with English skills. [28] presented a supportive robotic tutor that was used to teach language learning in a game like a scenario. A tutoring application was made with iCat Robot that was used for vocabulary learning. [11] also used the robots in home-based setup to teach languages. They compared robots teaching languages to audiotapes and books and showed that robots performed better, primarily due to their physical characteristics. Children preferred learning with the robot as compared to audiotapes. [18] performed a story-telling activity with robots as learning partners with preschool childrens to encourage language development. They designed a robot capable of narrating a story. As the robot narrates the story, a corresponding scene appears on the iPad. The story narration has three levels of difficulties. The purpose of this game was to check if the child was able to learn new vocabulary during the interaction. The results showed that children were able to learn new nouns, verbs and adjective through the narration.

In general, the use of agents/robot to support learning is a known process. However, questions have been raised with respect to sustaining long-term task interest and engagement during human agent interaction after a certain amount of time [16]. Researchers [24] have pointed that programming these robots with adaptive behaviours could lead to the solution of the mentioned problem. We believe that in order to implement adaptive social robots, we need to understand appropriate and effective adaptive behaviours in different domains to overcome these problems.

RESEARCH QUESTION

Our research questions were twofold. Firstly, we tried to understand about the current teaching practises used at the schools and how a humanoid robot can contribute. Secondly, we intended to understand the perspective of teachers on both existing and novel adaptation behaviours and social roles a robot can portray during both individual and group Children Robot Interactions (cHRI) in the classroom for language learning purposes. We also wanted to understand the effect of these adaptations on child's learning and overall engagement.

METHOD

Participant

We interviewed teachers from both primary and high schools in an urban city, where the first language of communication was English. All of these teachers had training and experience in teaching languages including French, Italian and English. We selected language teachers because we intend to implement an adaptive social robot that can support children with language learning and development. Eight female teachers participated in the study. Out of eight teachers, we had four teachers each from both primary and high school. The average age-range of teachers was between 25 and 45. The interview sessions were held at the corresponding schools and lasted for 30 minutes each. All of the interviews followed the same procedure in both schools.

Procedure

We began our study by distributing information sheets and consent forms to the teachers. Once teachers have completed and returned their consent forms, we gave them an introduction about the scope of the project and inquired in case each of them had any questions.

The study was later conducted with teachers during one-to-one interactions. Each study session lasted for 30 minutes and had three different steps: 1) a 5-minute video (<https://www.youtube.com/watch?v=ho9i1moUJos>) on HRI and SR followed by a 3-minute video (<https://www.youtube.com/watch?v=2STNYNF41k>) on NAO, 2) a 5-minute of interaction between teachers and NAO, 3) Interviews with teachers on how NAO can contribute towards language learning through adapting to children behaviours. The data collection was completed in two days. On the first day, we visited high school followed by our visit to the a primary school on the second day.

Firstly, we showed teachers a basic video on SR and HRI and NAO robot showing various capabilities such as gestures, object recognition, tactile sensors and speech recognition. We later asked teachers about possible confusions or questions on NAO and HRI in general. The purpose of showing these videos was to avoid an effect of previous and hypothetical knowledge about robots and also to overcome any fears that robots can takeover their jobs. Secondly, we asked teachers to interact with NAO in three different scenarios. In Scenario 1, teachers interacted with NAO to get familiar with its Speech Recognition capabilities. Teachers were asked to speak with NAO through asking the basic set of introductory questions such as: (*What is your name, How are you, How is your day progressing etc...*). Upon asking an unknown question, NAO repeated their question. In Scenario 2, teachers were able to interact with NAO through showing different emotions (*happy, sad, angry and neutral*) to understand about emotion detection capabilities. Teachers were asked to show emotions and NAO after detecting these emotions named the emotion. In Scenario 3, NAO showed different gestures (clapping, waving, bowing and dancing) upon teachers interactions. Teachers were asked to speak the gesture name to NAO and in response, NAO displayed the gesture. Lastly, we interviewed teachers in order to understand current practices followed by them at schools for language learning and how can NAO contribute towards enhancing child's learning. In addition, we asked teachers about their opinion on different adaptive behaviours and roles that a robot can display to influence child learning and long-term engagement.

Setup and Materials

We used NAO robot designed and developed by Aldebaran robotics. It is a humanoid robot measuring 58 cm in height. NAO is an interactive and adaptable robot partner. It provides researchers a platform to design various applications driven by their creativity and requirements.

We conducted our study in a quiet room at both primary and high schools during school time. We were provided with a small table and several chairs inside an empty classroom. The

NAO robot was placed in a sitting position in front of the participant teacher in order to get a the clear view of the participant's face for detecting emotions. The researcher sat in front of the participant and was involved in showing videos on the notebook and conducting interviews. The NAO robot autonomously generated behaviours for three different scenarios: *Speech recognition, emotions detection and displaying gestures*. The setup is shown in figure 1. We took consent from the teacher to use their picture in our article.

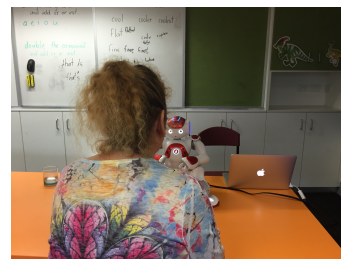


Figure 1. Setup: A teacher interacting with NAO.

NAO Robot and implementation of Scenarios

The NAO robot was autonomously programmed to display three different capabilities: Speech Recognition, detecting emotions and displaying gestures.

We used *google* speech to text API [32] to convert teacher's speech to text. The NAO robot was programmed to respond according to basic questions of the teachers. The response ranged from introducing itself to greeting each other. It was also able to repeat sentences of the teachers.

We also trained data on basic human emotions (Happy, sad, angry and neutral). We programmed NAO to capture user facial expressions and detect their emotions through using an algorithm [8]. The algorithm uses openCV library to localise the mouth area to detect emotion of a user. The image captured by NAO is re-sized to 28*10 pixel containing only person's mouth and surrounding areas. The image is then converted into grey-scale and flattened into a vector of length 280. A logistical regression programme then takes the vector and determines the emotional state of the user.

We implemented state of the art existing behaviours in NAO robot in *Choregraphe*. The gestures included bowing, clapping, touching, whipping, hugging and dancing. The *Choregraphe* programme was later used to generate python code.

Interview Questionnaire

The questionnaires involved understanding different language learning strategies adopted by the teachers and how NAO can adopt these strategies and contribute towards efficient learning. In addition, what are the novel adaptive behaviours and roles NAO should play to prolong children robot engagement? We understand that it is important to ask these questions because the generated data will give us directions to implement novel scenario and behaviours for future robots in an effective and appropriate way. It was also important to enable teachers interact with one of the Humanoid robots (NAO) in order to develop their understanding on robot's current capabilities. The questionnaire was designed according

Language	Teacher
French	P7,P4
Italian	P5,P6,P3,P8
English	P1,P2

Table 1. Teachers and Corresponding Languages

to the existing characteristics of an adaptive social robot [9]. Other questions were used from [30]. The list of other questions from our interviews are:

- I *What teaching approaches, resources and technologies do you use to teach oral and written language development?*
- II *How do you think a robot can contribute towards efficient language learning?*
- III *How do you want a robot to show different gestures during a one to one interaction?*
- IV *How do you want a robot to display a personality according to a child?*
- V *How do you want a robot to react to children emotions?*
- VI *What kind of role a robot should play to improve learning?*
- VII *How do you want a robot to store child's memory?*
- VIII *How to ensure long-term engagement between NAO and a child?*

Data Analysis

All of the interviews were audio recorded after receiving consent from the teachers. We performed content analysis [6] to analyse interviews communication. One of the authors listened to and transcribed the interviews. The author then noted possible themes and patterns in teachers responses. These patterns were later used to define main themes and sub-themes for this paper. Quotes were selected from the relevant responses.

RESULTS

We present the main themes resulting from the qualitative analysis performed on the interviewed data. First, we present teachers views on how NAO robot can contribute towards child's language learning and development. Second, we present about the behavioural and role adaptation for NAO according to teachers in Language language. We also present sub-themes with respect to adaptation in teacher's perspective. Thirdly, we present teacher's opinion on how NAO can maintain long-term engagement. To keep the identities anonymous, primary school teachers are labelled with the range of 1 to 8, where P1, P2, P3, P4 are from the primary school while P5, P6, P7, and P8 are from high school. Table 1 shows the languages taught by the corresponding participants.

Robots and its Contributions towards language learning

Teachers pointed out the robots can help with vocabulary learning, grammar, and correcting pronunciation. They can also play games with children on language learning. The

teachers also wanted children to practise their speaking skills with the robot. Some of the teachers reported:

We can possibly use a robot for word learning, comprehension and games based on questions and answers to motivate students to learn languages (P4).

The robot can speak a grammatically correct sentence and a child can repeat it (P2).

I think, the girls and the student will find it quite comfortable to interact with, they can practise their speaking skills (P6).

Teachers emphasised the benefits of programming robots that can help them with marking both objective and subjective assessments. They also referred to the benefit of having robots to fulfill drill types of tasks. For example, they stated that students usually keep asking same questions in the class. It can sometimes get frustrating for them but as the robot can repeat it as many times so it can be a great support in this regard.

Students normally ask same questions repeatedly, the robot can answer these questions in the classroom while I am working with other students (P8).

In addition, the teachers commented that it would be great if a robot can autonomously find learning material for them.

Robots with Gesture Adaptation

Teachers recognised gestures as one of the essential parts of language education. Robot depicting several appropriate gestures can be a handful and may lead towards efficient learning and engagement.

I am quite animated during my class. Gestures such as pointing, indicating, arm moments, bending, hands on heads works best with juniors (P5).

Teachers also mentioned a list of effective gestures to engage students in learning. It included: *pointing, arm moments, bending, hands on heads, greetings, hands up, hands down, use of finger to point the wrong answer, thumbs up, attention, listen please, finger going to the eye, saluting, throwing, dancing, and smiling.*

Teachers also mentioned that different gestures can be associated with different languages. They indicated that the gesture for greeting in the French language is different from other languages. The robot can have a different teaching style based on the language. One of the teachers was of the view that gestures can be culturally driven so if a robot needs to adapt accordingly. Cultural adaptation can also motivate child towards efficient learning.

We have a different way of saying good morning in french. We teach students regarding individual situations while teaching languages. The Robot can combine both gesture and speech capabilities to improve child's learning (P7).

Robots with Memory Adaptation

Teachers complemented memory adaptation as they stated in language learning recalling or remembering an experience can be a handful and can result in a positive effect on the

child's learning. They also mentioned that memory can be used as a great tool to show extra care towards a child and a child can be amused as well as motivated towards learning. One of the teachers reported:

We can show extra care through recalling previous interactions. If this can be made possible, children will not do the same mistake over and over again. Children will be forced to think about new ways of adaptation (P3).

One of the teachers also reported that memory can be a great tool to test student's performance. She reported:

Robot with child's memory can a great idea. If a student was able to say a certain word after two weeks, the robot can later ask the same question in the next days. It can, therefore, judge the child and also encourage through providing positive reinforcement (P6).

Robots with Emotional Adaptation

All of the teachers acknowledged the importance of recognising emotions through facial expressions. However, they were worried that facial expressions are not the only way to recognise emotions because they are not the only identifier of a child's mood or emotional state. Teachers focused on the importance of detecting the emotional state of the child through dialogue. They proposed a set of conversations that a robot should have in order to realise on the emotional state of the child.

The use of facial expression to detect emotion is important, However, we need the robot to sense mood through dialogue. For example, if a child is bored, then it can provide positive reinforcement or if the robot can detect anger, it can play a calming role (P3).

One of the teachers was of the view that in case, a child is not in a good mood on a particular day due to some unavoidable experience, then the robot should be able to judge it and perform certain steps to overcome or change the emotional state of the child.

The dialogue-based mood detection is more important than facial expression. The robot should react non-judgement and should also try to get them through a list of procedures and calm them accordingly (P5).

Teachers informed that robots having feelings will improve engagement as children will not only consider robot as the toy, however, they were concerned that extensive emotional adaptation might intimidate children of a certain age.

Robots with emotions makes them real and makes the student realise that robot has feelings and it is a real interaction. The robot understanding emotion can be intimated depending on the age of the student (P7).

Adaptive Social Roles for Robots

Teachers pointed various roles for the robot that it can display and adapt during children robot interaction. Table 2 shows a taxonomy of the frequency of various roles that in view of all the teachers can be appropriate or misleading for children. We coded these frequencies from teacher's statement.

Roles	Teachers in favour
Assertive	4
Funny	8
Encouraging	8
Persuasive or Motivating	8
Bribing	1
Dominant	1
Submissive	1
Democratic	5
Cooperative	8
Assistant or a Helper	8
Buddy or a Friend	8
Competitive	8
Calm and Compassionate	8

Table 2. Roles for a Robot in Language Learning

In general, teachers informed that robots need to adapt different role for the different scenario. Most of them said that robot needs to encourage and motivate children through persuasive dialogue. They also said Robot needs to find a balance between a submissive and dominant role and should rather play a democratic role. One of the teachers, however, was also of the opinion that dominance is important in order to be taken seriously.

If a student is hesitant to do something, then Robot should be able to persuade him/her through positive reinforcement (P5).

Student would like the idea that the robot adapts and play a buddy role in the classroom. It can be a motivator for children to learn languages (P7).

Robot displaying a dominant role can be overwhelming for the child (P2).

All of the teachers pointed attention towards the use of a robot with a good sense of humour as learning a language can get boring at times. The robot needs to play funny roles in order to motivate children and keep them engaged during the learning process.

Robot needs to act funny and tell jokes to students, for example, when explaining the meaning of a word in a vocabulary learning exercise. This will keep the child engaged and motivated throughout the process.

Teachers considered calm and compassionate roles to be best suited for the case when the child is making repetitive mistakes. The robot needs to first recognise the repeated mistake and adapt through changing the dialogue or its tone in real time.

You need to play a calming or soothing role through the use of dialogue and gesture in order to motivate and help the child improve learning (P8).

Robots with Personality Adaptation

Teachers were of the opinion that it is not necessary to adapt according to the general definition of personality (extrovert or introvert) of the child. They commented that child's person-

ality might vary on the day according to the set of events that might have happened during the school time. The robot, on the other hand, should be able to recognise child's personality through dialogue. Teachers reported as:

Personality adaptation through dialogue can be an efficient way, in a personalised interaction, personality can be a hand-ful, for example, the robot needs to show patience in case of an introvert child (P5).

We need to detect the personality of the student, the person-ality can change over time so we need a mechanism to detect the personality. We can detect that through dialogue, for in-stance, we can detect hesitation then encourage the student through rewards or positive gestures (P6).

Teachers also directed towards finding the balance between an extrovert and introvert personality. A teacher reported:

A balance between an extrovert and introvert is required. Perhaps a more outgoing one as it would probably encourage more interaction (P1).

Robots for Long-term Engagement

Teachers expressed concerns about long-term engagement between children and the robot. They were certain that in order to achieve sustainable engagement, they should be equally involved in preparing robot interactions with the children. They were afraid that if teachers are not given control of the robot, it could very well happen that the robot stays alone at the corner in a classroom. In the words of one teacher:

If the teachers know how to control and use the robot, then we would be able to achieve long-term engagement (P5).

Another teacher pointed out that teacher should be provided with an interface that enables them to update or change lessons over the period of time.

Teachers can keep the robot involved, so that it not just at the corner and always in the loop, we need to use the robot as a part of a lesson (P6).

However, another teacher also mentioned that if there are a variety of programs and the systems updates itself and also adapts to a child's need, it would be possible to achieve long-term engagement.

You should have a variety of programs so that the robot is not doing the same thing over and over again. The robot should mimic what a human would do at a certain moment (P1).

DISCUSSION

In this study, we have learnt a number of lessons and take-aways for HRI researchers. The key lessons we learned about implementing adaptive social robots in language education, as reflected in the teacher's feedback are as follow:

1. The robot should be able to answer repeatedly asked questions in the classroom: There is a need to implement a mechanism in which a robot can respond to repeatedly asked questions during one-to-one or group interactions with children. In order to achieve it, we need to implement ways for a

robot to perform memory adaptations. Chang and colleagues [3] have described various characteristics of robots that can be important during language learning. Repeatability was described as one of the most appropriate features for language education.

2. We need to design dialogue based adaptation mechanism in order to adapt to children emotions and personality in real-time: Human emotions and personality are correlated to each other. The current emotional state or mood of humans can influence the portrayal of personality. Therefore, as indicated by the teachers, it is significant to design and implement a real-time adaptation mechanism for a robot to detect user's personality based on its emotional state. Previous studies [13][23] have shown that if a robot can adapt according to the personality of the user, it can positively influence learning. However, most of the studies in which a robot is able to adapt users personality are conducted through asking participants to complete a questionnaire in order to detect their personality (extrovert or introvert). Therefore, it remains an open research question on how to detect personality in real-time. In addition, another possible strategy is to use a broad categorization of emotions that move beyond a simple extrovert and introvert identification. On the other hand, Mood or emotion adaptations based on dialogue have been studied in human-computer interaction [27]. However, we find fewer work on dialogue-based emotion adaptation and its effect on user perception and performance for educational robots and HRI in general.

3. The selection of robot role's during children robot interaction in real-time can be based on memory adaptation: Most of the teachers emphasised on performing real-time role adaptation. One of the suggested ways was through the use of memory during children robot interaction. In literature, we find a number of studies where robots have played different roles of a friend [7], competitor or cooperative [20], and persuasive [4]. All of these studies have shown positive results with respect to robot's effect on user perception, engagement or learning. However, there is a need for designing mechanism such that a robot can adapt its role in real time according to a given situation. In addition, the use of memory to perform role based adaptation still remains an open question in HRI.

4. Robots keeping track of a child's memory can in-turn motivate children: Teachers mentioned that children make mistakes regularly. The robot with a capability to show an adaptive behaviour through the use of recalling previously made mistakes can motivate children learning. As teachers mentioned that it can enable children to think new ways of adaptations in order to impress or outsmart the robot. Literature in HRI has also emphasised on the significance of memory adaptation. In a recent survey conducted on social robots for long-term engagement, [21] has mentioned memory based adaptation as one of the unexplored areas in HRI. Leite and Colleagues [21] have conjectured that possessing a memory can make social robots more flexible and personalised to particular users. Recently, researchers have also shown that robots with memory can affect user performance

as it enhances their likability and empathy [12]. Therefore, teachers opinion are in line with the current research recommendations in HRI.

5. Culture-based adaptation can be significant during language learning tasks: Teachers explained that the significance of gestures during language learning. They also mentioned that robot should adapt when teaching different languages because gestures are culturally driven for different languages. We find less work on Cultural based adaptation in HRI. Studies have been conducted to show the significance of culture during HRI. Researchers [33] have indicated the understanding of gestures is dependent on the culture and even within one culture, interpretations can differ for different situations. In addition, it has also been described that different individuals from different cultures perceive personality of the robot differently [15]. Therefore, it is important to study the effect of cultural adaptations on user's engagement and performance during HRI.

6. Consider designing an easy to use interface for teachers to update new lessons for long-term Engagement: Teachers emphasised the importance of their involvement in order to keep the robot engaged and involved during the learning process for long time. In order to keep teachers involved, it is important to design interfaces that allows them to manage robots. The robot needs to be adaptive, however, the content and curriculum need to be revised or changed after a certain period, which can be done by the teacher only if he/she has appropriate control. In literature, [3] has described that the robot needs to be flexible enough to allow teachers to adjust and design appropriate robot-supported instructional activities for relevant teaching and learning requirements. In addition, Orlando et. al. [26] has showed that teachers today have confidence with technology and also possess a diverse range of technology expertise. Therefore, research needs to be conducted on the development of such interfaces that are easy to use for teachers.

LIMITATIONS, CONCLUSION AND FUTURE WORK

We didn't conduct the study longitudinally so we were not able to overcome the biases of the teachers. However, during the study, we tried to overcome pre-existing biases through the use of videos and the actual presence of the robot. The teachers agreed to have understood the capabilities of the robot and confirmed that their responses won't be driven based on any biases or past experience. Another limitation to our work is that the teachers were able to interact with limited capabilities (e.g. the robot only responded to limited questions, only recognised basic emotions, and displayed few gestures). We also understand if teachers were able to interact with NAO that showed more capabilities, we would have received more classified information from the teacher. Another argued limitation of our study can be the total number of participants. However, one of the previous studies [31] conducted with teachers on their views on robots in education also had the equal number of participants. In addition, our participants were specifically language teachers therefore we can interpret our results with some reliability.

In this paper, we presented our results on teacher's opinion on how robots can contribute towards language learning with children through performing a series of adaptations. The adaptations can be based on memory, emotions, and personality of the child. In addition, we show that there is a need to implement easy to use interfaces for teachers to enable them to upload new content for the robot. This can lead towards long term engagement of robots in education.

In future, we will be conducting a study with children (10-12 year old) at schools about their views on robot's that can adapt to their actions. We will be looking to conduct our study in three different steps: 1) an adaptive robot plays a snakes and ladders game for vocabulary learning [1] with the child individually and in pairs, 2) we will later invite children in groups to discuss robots with adaptive abilities and what are their views on them and 3) respond to individual asked interview questions. We will be recording videos of all these sessions with children. We will be looking to analyse videos in order to get data on how children react with robots in different scenarios during game-play and we will use this data to identify a list of efficient and appropriate children-robot interactions.

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